

# CSE 385-01 (#49225): Analysis of Algorithms: Honors, Spring 2026

**Textbooks.** These are recommended, but not required.

1. Thomas Cormen, Charles Leiserson, Ronald Rivest, and Clifford Stein. *Introduction to Algorithms* (3rd/4th Edition), MIT Press, 2009/2022.
2. Steven Skiena. *The Algorithm Design Manual* (2nd/3rd Edition), Springer, 2008/2020.
3. Sanjoy Dasgupta, Christos Papadimitriou, and Umesh Vazirani. *Algorithms* (1st Edition), McGraw-Hill, 2006.
4. Jon Kleinberg and Éva Tardos. *Algorithm Design* (1st Edition), Addison Wesley, 2005.
5. Pramod Ganapathi. *Mathematical and Algorithmic Puzzles*, [Free online version](#), 2021.

## Reading Material.

| Lecture   | Topic                                                                                 | Reading Material                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
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| Lecture 1 | Introduction                                                                          | <ul style="list-style-type: none"><li>• <b>[optional]</b> <i>Algorithmic Puzzles</i> by Anany Levitin and Maria Levitin, Oxford University Press, 2011.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                       |
| Lecture 2 | Asymptotic Analysis of Algorithms                                                     | <ul style="list-style-type: none"><li>• VIII Appendix: Mathematical Background, <i>Introduction to Algorithms</i> (4th Edition) by Cormen et al.</li><li>• Chapter 3 (Characterizing Running Times), <i>Introduction to Algorithms</i> (4th Edition) by Cormen et al.</li></ul>                                                                                                                                                                                                                                                                         |
| Lecture 3 | Correctness of Algorithms: Insertion Sort and Selection Sort                          | <ul style="list-style-type: none"><li>• Chapter 2 (Getting Started), Section 2.1 (Insertion Sort), <i>Introduction to Algorithms</i> (4th Edition) by Cormen et al.</li><li>• Chapter 2 (Getting Started), Section 2.2 (Analyzing Algorithms), <i>Introduction to Algorithms</i> (4th Edition) by Cormen et al.</li></ul>                                                                                                                                                                                                                               |
| Lecture 4 | Correctness of Algorithms / Divide-and-Conquer Algorithms: Merge Sort                 | <ul style="list-style-type: none"><li>• Chapter 2 (Getting Started), Section 2.3 (Designing Algorithms), <i>Introduction to Algorithms</i> (4th Edition) by Cormen et al.</li></ul>                                                                                                                                                                                                                                                                                                                                                                     |
| Lecture 5 | Divide-and-Conquer Algorithms: Integer Multiplication & Karatsuba's Algorithm         | <ul style="list-style-type: none"><li>• Chapter 2 (Divide-and-Conquer Algorithms), Section 2.1 (Multiplication), <i>Algorithms</i> (1st Edition) by Dasgupta et al.</li><li>• <b>[optional]</b> Anatolii A. Karatsuba, <a href="#">“The Complexity of Computations”</a>, Proceedings of the <i>Steklov Institute of Mathematics</i>, 211:169-183, 1995.</li></ul>                                                                                                                                                                                       |
| Lecture 6 | Divide-and-Conquer Algorithms: Matrix Multiplication & Strassen's Algorithm           | <ul style="list-style-type: none"><li>• Chapter 4 (Divide-and-Conquer), Section 4.2 (Strassen's Algorithm for Matrix Multiplication), <i>Introduction to Algorithms</i> (4th Edition) by Cormen et al.</li><li>• <b>[optional]</b> Chapter 9 (Algebraic and Numeric Algorithms), Section 9.5.2 (Strassen's Algorithm), <i>Introduction to Algorithms - A Creative Approach</i> (1st Edition) by Udi Manber.</li><li>• Volker Strassen, <a href="#">“Gaussian Elimination is not Optimal”</a>, <i>Numerische Mathematik</i>, 13:354-356, 1969.</li></ul> |
| Lecture 7 | Divide-and-Conquer Algorithms: Polynomial Multiplication & the Fast Fourier Transform | <ul style="list-style-type: none"><li>• Chapter 30 (Polynomials and the FFT), <i>Introduction to Algorithms</i> (4th Edition) by Cormen et al.</li><li>• Chapter 2 (Divide-and-Conquer Algorithms), Section 2.6 (The Fast Fourier Transform), <i>Algorithms</i> (1st Edition) by Dasgupta et al.</li></ul>                                                                                                                                                                                                                                              |

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|            |                                                       | <ul style="list-style-type: none"> <li>James W. Cooley and John W. Tukey, <a href="#">“An Algorithm for the Machine Calculation of Complex Fourier Series”</a>, Mathematics of Computation, 19(90):297–301, 1965.</li> <li>[fun] Thomas S. Huang, <a href="#">“How the Fast Fourier Transform Got its Name”</a>, Computer, 4(3), page 15, 1971.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Lecture 8  | Divide-and-Conquer Algorithms: The Master Theorem     | <ul style="list-style-type: none"> <li>Chapter 4 (Divide-and-Conquer), Section 4.5 (The Master Method for Solving Recurrences), <i>Introduction to Algorithms</i> (4th Edition) by Cormen et al.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Lecture 9  | Divide-and-Conquer Algorithms: Akra-Bazzi Recurrences | <ul style="list-style-type: none"> <li>Chapter 9 (Medians and Order Statistics), Section 9.3 (Selection in Worst-case Linear Time), <i>Introduction to Algorithms</i> (3rd Edition) by Cormen et al.</li> <li>Mohamad Akra and Louay Bazzi, <a href="#">“On the Solution of Linear Recurrence Equations”</a>, Computational Optimization and Applications, 10(2):195–210, 1998.</li> <li>Tom Leighton, <a href="#">“Notes on Better Master Theorems for Divide-and-Conquer Recurrences”</a>, 1996.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Lecture 10 | Deterministic Quicksort and its Average Case Analysis | <ul style="list-style-type: none"> <li>Chapter 7 (Quicksort), Section 7.1 (Description of Quicksort), <i>Introduction to Algorithms</i> (3rd Edition) by Cormen et al.</li> <li>Chapter 7 (Quicksort), Section 7.2 (Performance of Quicksort), <i>Introduction to Algorithms</i> (3rd Edition) by Cormen et al.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Lecture 11 | Greedy Algorithms                                     | <ul style="list-style-type: none"> <li>Chapter 16 (Greedy Algorithms), Section 16.1 (An Activity Selection Problem), <i>Introduction to Algorithms</i> (3rd Edition) by Cormen et al.</li> <li>Chapter 16 (Greedy Algorithms), Section 16.2 (Elements of the Greedy Strategy), <i>Introduction to Algorithms</i> (3rd Edition) by Cormen et al.</li> <li>Chapter 23 (Minimum Spanning Trees), Section 23.1 (Growing a Minimum Spanning Tree), <i>Introduction to Algorithms</i> (3rd Edition) by Cormen et al.</li> <li>Chapter 23 (Minimum Spanning Trees), Section 23.2 (The Algorithms of Kruskal and Prim), <i>Introduction to Algorithms</i> (3rd Edition) by Cormen et al.</li> <li>Chapter 24 (Single-Source Shortest Paths), Section 24.3 (Dijkstra's Algorithm), <i>Introduction to Algorithms</i> (3rd Edition) by Cormen et al.</li> <li>Mo Chen, Rezaul Chowdhury, Vijaya Ramachandran, David Lan Roche, and Lingling Tong, <a href="#">“Priority Queues and Dijkstra's Algorithm”</a>, UT Austin, Department of Computer Sciences, Technical Report TR-07-54, Oct. 2007.</li> </ul> |
| Lecture 12 | More Greedy: Network Flow and Bipartite Matchin       | <ul style="list-style-type: none"> <li>Chapter 26 (Maximum Flow), Section 26.1 (Flow networks), <i>Introduction to Algorithms</i> (3rd Edition) by Cormen et al.</li> <li>Chapter 26 (Maximum Flow), Section 26.2 (The Ford-Fulkerson method), <i>Introduction to Algorithms</i> (3rd Edition) by Cormen et al.</li> <li>Chapter 26 (Maximum Flow), Section 26.3 (Maximum bipartite matching), <i>Introduction to Algorithms</i> (3rd Edition) by Cormen et al.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

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| Lecture<br>13 | Heaps and Heapsort                            | <ul style="list-style-type: none"> <li>Chapter 6 (Heapsort), Section 6.1 (Heaps), <i>Introduction to Algorithms</i> (3rd Edition) by Cormen et al.</li> <li>Chapter 6 (Heapsort), Section 6.2 (Maintaining the Heap Property), <i>Introduction to Algorithms</i> (3rd Edition) by Cormen et al.</li> <li>Chapter 6 (Heapsort), Section 6.3 (Building a Heap), <i>Introduction to Algorithms</i> (3rd Edition) by Cormen et al.</li> <li>Chapter 6 (Heapsort), Section 6.4 (The Heapsort Algorithm), <i>Introduction to Algorithms</i> (3rd Edition) by Cormen et al.</li> <li>Chapter 6 (Heapsort), Section 6.5 (Priority Queues), <i>Introduction to Algorithms</i> (3rd Edition) by Cormen et al.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Lecture<br>14 | Backtracking                                  | <ul style="list-style-type: none"> <li>Chapter 2 (Backtracking), <a href="#"><i>Algorithms</i></a> (1st Edition) by Jeff Erickson</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Lecture<br>15 | Dynamic Programming                           | <ul style="list-style-type: none"> <li>Chapter 15 (Dynamic Programming), Section 15.1 (Rod Cutting), <i>Introduction to Algorithms</i> (3rd Edition) by Cormen et al.</li> <li>Chapter 15 (Dynamic Programming), Section 15.2 (Matrix-chain Multiplication), <i>Introduction to Algorithms</i> (3rd Edition) by Cormen et al.</li> <li>Rezaul Chowdhury, Pramod Ganapathi, Steven Tschudi, Jesmin Jahan Tithi, Charles Bachmeier, Bradley C. Kuszmaul, Charles E. Leiserson, Armando Solar-Lezama, and Yuan Tang, <a href="#">“AutoGen: Automatic Discovery of Efficient Recursive Divide-&amp;Conquer Algorithms for Solving Dynamic Programming Problems”</a>, ACM Transactions on Parallel Computing, 4(1):4, 2017.</li> <li>Chapter 15 (Dynamic Programming), Section 15.3 (Elements of Dynamic Programming), <i>Introduction to Algorithms</i> (3rd Edition) by Cormen et al.</li> <li>Chapter 15 (Dynamic Programming), Section 15.4 (Longest Common Subsequence), <i>Introduction to Algorithms</i> (3rd Edition) by Cormen et al.</li> <li>Chapter 15 (Dynamic Programming), Section 15.5 (Optimal Binary Search Trees), <i>Introduction to Algorithms</i> (3rd Edition) by Cormen et al.</li> <li>Rezaul Chowdhury and Vijaya Ramachandran, <a href="#">“Cache-oblivious Dynamic Programming”</a>, Proceedings of the 17th annual ACM-SIAM symposium on Discrete algorithm, pp. 591-600, 2006.</li> </ul> |
| Lecture<br>16 | More Shortest Paths and Other Graph Algorithm | <ul style="list-style-type: none"> <li>Chapter 22 (Elementary Graph Algorithms), Section 22.2 (Breadth-first search), <i>Introduction to Algorithms</i> (3rd Edition) by Cormen et al.</li> <li>Chapter 22 (Elementary Graph Algorithms), Section 22.3 (Depth-first search), <i>Introduction to Algorithms</i> (3rd Edition) by Cormen et al.</li> <li>Chapter 22 (Elementary Graph Algorithms), Section 22.4 (Topological sort), <i>Introduction to Algorithms</i> (3rd Edition) by Cormen et al.</li> <li>Chapter 22 (Elementary Graph Algorithms), Section 22.5 (Strongly connected components), <i>Introduction to Algorithms</i> (3rd Edition) by Cormen et al.</li> <li>Chapter 24 (Single-Source Shortest Paths), Section 24.1 (The Bellman-Ford algorithm), <i>Introduction to Algorithms</i> (3rd Edition) by Cormen et al.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

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|            |                                                   | <ul style="list-style-type: none"> <li>Chapter 24 (Single-Source Shortest Paths), Section 24.2 (Single-source shortest paths in directed acyclic graphs), <i>Introduction to Algorithms</i> (3rd Edition) by Cormen et al.</li> <li>Chapter 25 (All-Pairs Shortest Paths), Section 25.1 (Shortest paths and matrix multiplication), <i>Introduction to Algorithms</i> (3rd Edition) by Cormen et al.</li> <li>Chapter 25 (All-Pairs Shortest Paths), Section 25.2 (The Floyd-Warshall algorithm), <i>Introduction to Algorithms</i> (3rd Edition) by Cormen et al.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Lecture 17 | Amortized Analysis                                | <ul style="list-style-type: none"> <li>Chapter 17 (Amortized Analysis), <i>Introduction to Algorithms</i> (3rd Edition) by Cormen et al.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Lecture 18 | Binomial Heaps                                    | <ul style="list-style-type: none"> <li>Jean Vuillemin, <a href="#">“A data structure for manipulating priority queues”</a>, Communications of the ACM, 21(4):309-315, 1978.</li> <li><b>[optional]</b> Chapter 8 (Binomial Heaps), <i>The Design and Analysis of Algorithms</i> (1992) by Dexter Kozen.</li> <li><b>[optional]</b> Chapter 19 (Binomial Heaps), <i>Introduction to Algorithms</i> (2nd Edition) by Cormen et al.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Lecture 19 | Dijkstra's SSSP & Fibonacci Heaps                 | <ul style="list-style-type: none"> <li>Chapter 19 (Fibonacci Heaps), <i>Introduction to Algorithms</i> (3rd Edition) by Cormen et al.</li> <li>Michael L. Fredman and Robert E. Tarjan, <a href="#">“Fibonacci heaps and their uses in improved network optimization algorithms”</a>, Journal of the ACM, 34(3):596-615, 1987.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Lecture 20 | Randomized Algorithms and High Probability Bounds | <ul style="list-style-type: none"> <li>Appendix C (Counting and Probability), <i>Introduction to Algorithms</i> (3rd Edition) by Cormen et al.</li> <li><b>[optional]</b> Chapter 6 (Algorithms Involving Sequences and Sets), Section 6.9.2 (A Coloring Problem), <i>Introduction to Algorithms - A Creative Approach</i> (1st Edition) by Udi Manber.</li> <li><b>[optional]</b> Chapter 1 (Introduction), Section 1.1 (A Min-Cut Algorithm), <i>Randomized Algorithms</i> (1st Edition) by Rajeev Motwani and Prabhakar Raghavan.</li> <li><b>[optional]</b> Torben Hagerup and Christine Rüb, <a href="#">“A Guided Tour of Chernoff Bounds”</a>, Information Processing Letters, 33(6), pp. 305-308, 1990.</li> <li><b>[optional]</b> Chapter 7 (Quicksort), Section 7.4 (Analysis of Quicksort), <i>Introduction to Algorithms</i> (3rd Edition) by Cormen et al.</li> <li>William Pugh, <a href="#">“Skip lists: a probabilistic alternative to balanced trees”</a>, Communications of the ACM, 33(6):668-676, 1990.</li> </ul> |
| Lecture 21 | Reductions and NP-Completeness                    | <ul style="list-style-type: none"> <li>Chapter 8 (NP-Complete Problems), <i>Algorithms</i> (1st Edition) by Dasgupta et al.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Lecture 22 | Approximation Algorithms                          | <ul style="list-style-type: none"> <li>Chapter 35 (Approximation Algorithms), <i>Introduction to Algorithms</i> (3rd Edition) by Cormen et al.</li> <li>Chapter 35 (Approximation Algorithms), Section 35.1 (The Vertex-Cover Problem), <i>Introduction to Algorithms</i> (3rd Edition) by Cormen et al.</li> <li>Chapter 35 (Approximation Algorithms), Section 35.2 (The Traveling-Salesman Problem), <i>Introduction to Algorithms</i> (3rd Edition) by Cormen et al.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

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|            |                                                              | <ul style="list-style-type: none"> <li>• Chapter 35 (Approximation Algorithms), Section 35.2.1 (The Traveling-Salesman Problem with the Triangle Inequality), <i>Introduction to Algorithms</i> (3rd Edition) by Cormen et al.</li> <li>• Chapter 35 (Approximation Algorithms), Section 35.2.2 (The General Traveling-Salesman Problem), <i>Introduction to Algorithms</i> (3rd Edition) by Cormen et al.</li> <li>• Chapter 35 (Approximation Algorithms), Section 35.3 (The Set-Covering Problem), <i>Introduction to Algorithms</i> (3rd Edition) by Cormen et al.</li> <li>• Chapter 35 (Approximation Algorithms), Section 35.5 (The Subset-Sum Problem), <i>Introduction to Algorithms</i> (3rd Edition) by Cormen et al.</li> </ul>                                                       |
| Lecture 23 | Analyzing Parallel Algorithms                                | <ul style="list-style-type: none"> <li>• <b>[optional]</b> Chapter 5 (Analytical Modeling of Parallel Programs), <i>Introduction to Parallel Computing</i> (2nd Edition) by Grama et al.</li> <li>• Chapter 27 (Multithreaded Algorithms), <i>Introduction to Algorithms</i> (3rd Edition) by Cormen et al.</li> <li>• <b>[optional]</b> Gordon E. Moore, <a href="#">“Cramming more components onto integrated circuits”</a>, <i>Electronics</i>, 38(8), pp. 114-117, April 19, 1965.</li> <li>• Gene Amdahl, <a href="#">“Validity of the Single Processor Approach to Achieving Large Scale Computing Capabilities”</a>, Reprinted from the proceedings of the <i>Spring Joint Computer Conference</i>, AIFPS vol 30, pp. 483-485, 1967.</li> </ul>                                            |
| Lecture 24 | Analyzing I/O and Cache Performance                          | <ul style="list-style-type: none"> <li>• Alok Aggarwal and Jeffrey S. Vitter, <a href="#">“The input/output complexity of sorting and related problems”</a>, <i>Communications of the ACM</i>, 31(9), pp. 1116-1127, 1988.</li> <li>• Matteo Frigo, Charles E. Leiserson, Harald Prokop, and Sridhar Ramachandran, <a href="#">“Cache-Oblivious Algorithms”</a>, <i>Proceedings of the 40th Annual Symposium on Foundations of Computer Science (FOCS)</i>, pp. 285-298, 1999.</li> <li>• Jeffrey S. Vitter, <a href="#">“Algorithms and Data Structures for External Memory”</a>, <i>Series on Foundations and Trends in Theoretical Computer Science</i>, Now Publishers, Hanover, MA, 2008.</li> </ul>                                                                                         |
| Lecture 25 | The Inverse Ackermann Function, Union-Find, and Partial Sums | <ul style="list-style-type: none"> <li>• Chapter 21 (Data Structures for Disjoint Sets), <i>Introduction to Algorithms</i> (3rd Edition) by Cormen et al.</li> <li>• Raimund Seidel and Micha Sharir, <a href="#">“Top-Down Analysis of Path Compression”</a>, <i>SIAM Journal on Computing</i>, 34(3):515-525, 2005.</li> <li>• <b>[optional]</b> Andrew C. Yao, <a href="#">“Space-Time Tradeoff for Answering Range Queries”</a>, <i>Proceedings of the 14th Annual ACM Symposium on Theory of Computing (STOC)</i>, pp. 128-136, 1982.</li> <li>• <b>[optional]</b> Bernard Chazelle and Burton Rosenberg, <a href="#">“Computing Partial Sums in Multidimensional Arrays”</a>, <i>Proceedings of the 5th Annual Symposium on Computational Geometry (SCG)</i>, pp. 131-139, 1989.</li> </ul> |
| Lecture 26 | Generating Functions                                         | <ul style="list-style-type: none"> <li>• <b>[optional]</b> Chapter 7 (Advanced Counting Techniques), Section 7.2 (Solving Linear Recurrence Relations), <i>Discrete Mathematics and its Applications</i> (6th Edition) by Kenneth Rosen.</li> <li>• <b>[optional]</b> Chapter 7 (Generating Functions), <i>Concrete Mathematics</i> (2nd Edition) by Ronald Graham, Donald Knuth,</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                      |

and Oren Patashnik.

- **[optional]** Chapter 10 (Ordinary Generating Functions), Section 10.3 (Manipulating Generating Functions), Example 10.12 (The Average Time for Quicksort), *Foundations of Combinatorics with Applications* by Edward A. Bender and S. Gill Williamson.